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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

7187312255

ENVIRONMENTAL MANAGEMENT **5014/23**

Alternative to Coursework **May/June 2010**

1 hour 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Ruler

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
Study the appropriate Source materials before you start to write your answers.
Credit will be given for appropriate selection and use of data in your answers and for relevant interpretation of these data. Suggestions for data sources are given in some questions.
You may use the source data to draw diagrams and graphs or to do calculations to illustrate your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use

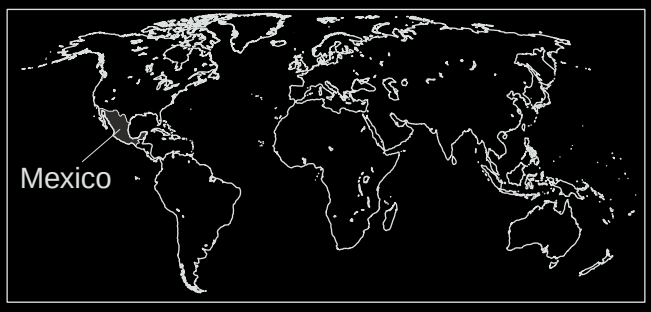
This document consists of **14** printed pages and **2** blank pages.

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Mexico



- Area of Mexico: 1 972 550 sq km
- Population: 115 million
- Children per woman: 2.34
- Life expectancy at birth: 76 years
- Currency: Mexican pesos (11.0 pesos = 1 US dollar)
- Languages: Spanish, local languages
- Climate: varies from wet tropical to desert
- Terrain: high, rugged mountains; coastal plains; high plateaus; desert
- Main exports: manufactured goods, oil and oil products, silver, fruits, vegetables, coffee and cotton

Mexico has a free market economy that depends on modern industries, agriculture and tourism. There are abundant reserves of oil, natural gas and minerals. Social concerns include low wages and underemployment, especially in the southern states such as Chiapas and Tabasco. Other problems include rural to urban migration, shortage of clean drinking water, deforestation and desertification.

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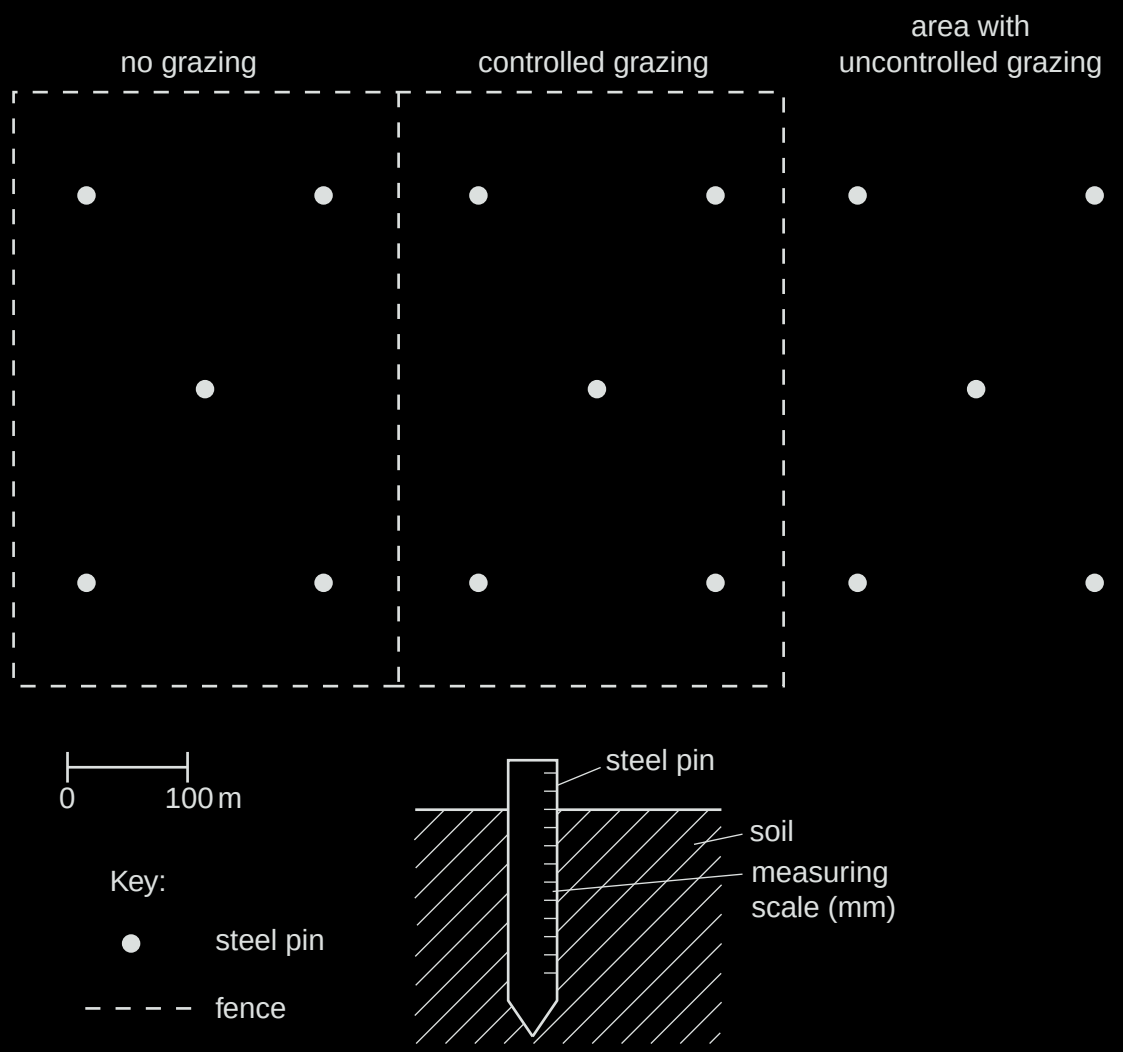


Fig. 1.1

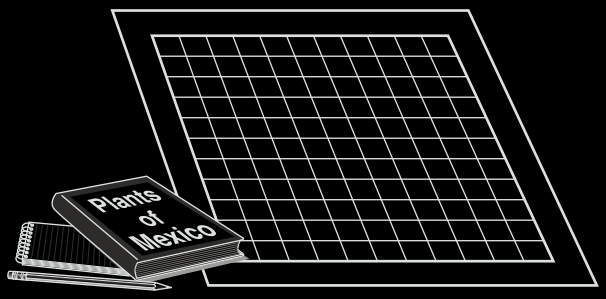
(i) Suggest why five steel pins were placed in the soil in each experimental area.

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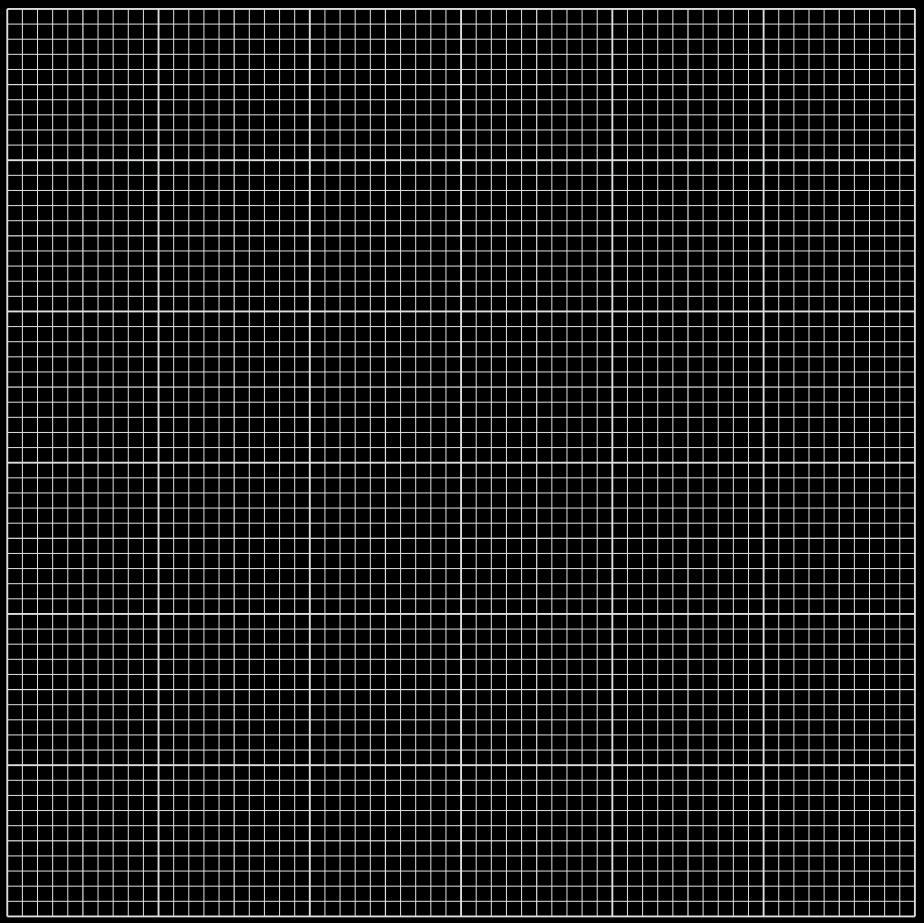
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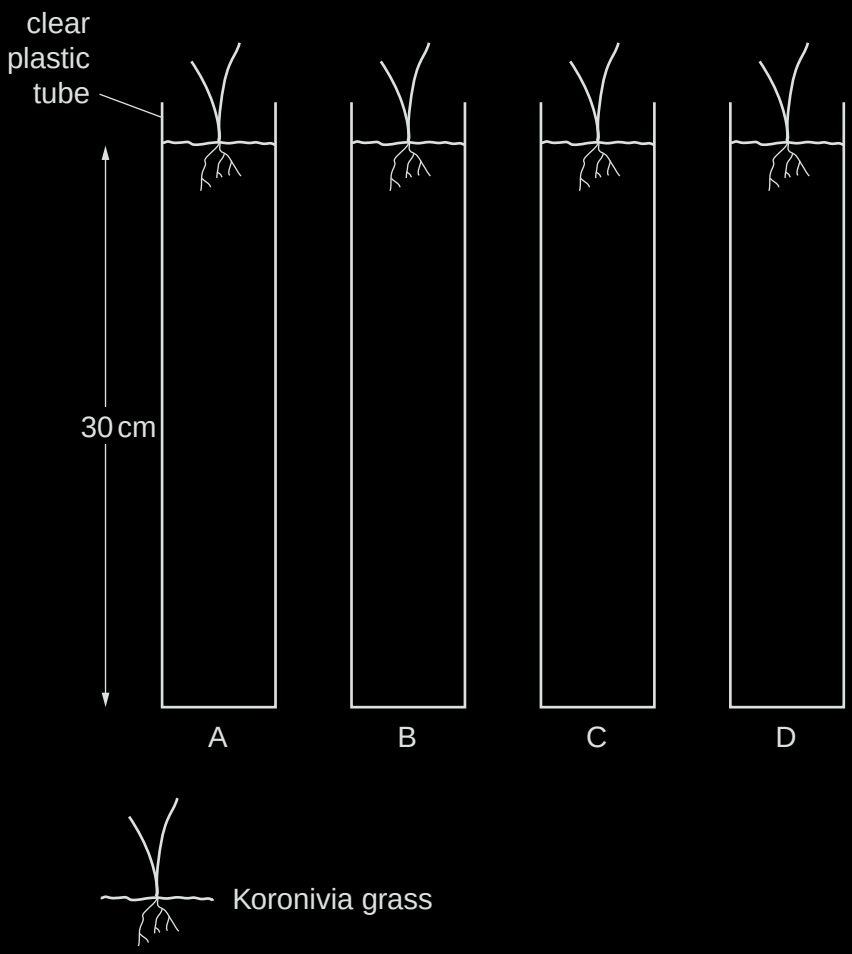


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	increase in length of grass / cm			
days after planting	tube A control soil	tube B polluted soil	tube C polluted soil	tube D polluted soil
0	0	0	0	0
4	3	2	1	2
8	8	5	6	6
12	15	15	16	14
16	24	28	29	27
20	36	40	42	41
24	48	54	56	53

Fig. 2.1

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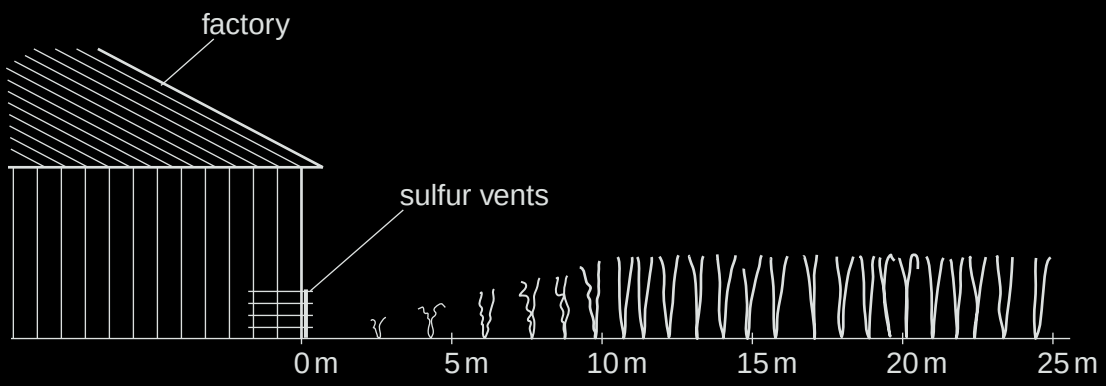


Fig. 2.2

(i) Does sulfur dioxide alter plant growth? Describe the evidence shown in Fig. 2.2.

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..... [2]

(ii) When sulfur dioxide is added to water in the air it forms an acid.

Name the acid formed.

..... [1]

(iii) Describe the effects of this acid on the vegetation and soil.

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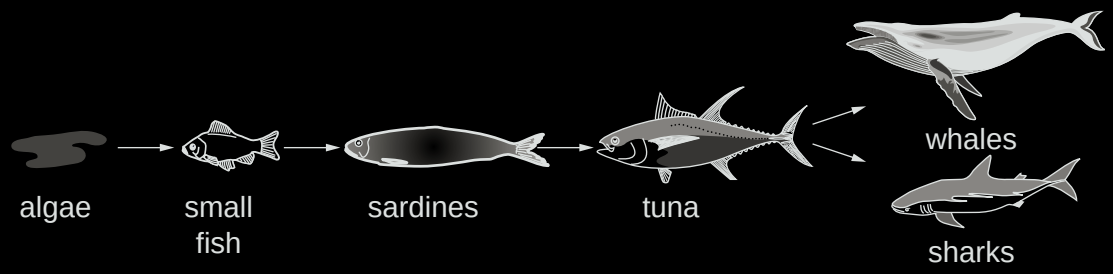


Fig. 3.1

Suggest likely effects on the food chain, shown in Fig. 3.1, if tuna species become rare due to overfishing.

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..... [2]

(d) Some sports fishermen are willing to pay millions of pesos to catch large fish such as tuna and marlin. If the fish stocks collapse local fishermen cannot earn money either from catching fish or taking sports fishermen to sea.

Suggest plans for sustainable sports fishing and sustainable fishing for food.

sports fishing .....

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fishing for food .....

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[6]

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